Milestone 10 | Streamlit + Plotly | Hydration Simulator | Batch Engine | SQLite Persistence

Milestone 10: Advanced Interactive Material Studio

Material Behavior Prediction • 8-Tab Engineering Decision Support & Simulation Platform

1. Overview of Interactive Enhancements

The application has been elevated into a comprehensive, portfolio-grade **Material Behavior Prediction & Mix Design Studio**. Built with **Streamlit** and **Plotly**, it enables materials scientists and civil engineers to run real-time simulations, stress-test formulations, perform sensitivity sweeps, assess carbon footprints, and batch-process laboratory CSV data.

The 8 Interactive Engineering Modules in the Studio

Tab	Module Name	Interactive Capabilities
Tab 1	**Hydration Kinetics**	Interactive 365-day strength progression curve (Plotly log-scaled) with active day pin, 7d/28d hydration ratio, and stripping benchmarks.
Tab 2	**Sensitivity Simulator**	"What-If" parameter sweep (50 points) across water, cement, superplasticizer, or SCMs with local marginal gain calculation (ΔMPa / Δunit).
Tab 3	**Abrams' Law Explorer**	Interactive scatter plot of 1,030 historical tests with hover tooltips, curing age filters, and candidate formulation overlay.
Tab 4	**Carbon & SCMs**	Embodied carbon breakdown donut chart, CO&sub2; reduction vs. pure OPC baseline, equivalent mature tree absorption metric, and eco-rating badges.
Tab 5	**Mix Comparator**	Head-to-head performance analysis (Mix A vs Mix B) with differential cards and grouped multi-metric bar charts.
Tab 6	**Recipe & Mass**	Volumetric mass distribution donut chart and batch weight ticket per 1 m^3.
Tab 7	**Batch CSV Predictor**	Drag-and-drop CSV uploader for multi-mix batch inference with distribution histogram and one-click enriched CSV download.
Tab 8	**Database Archive**	Searchable, sortable historical trial registry backed by SQLite with complete CSV export.

2. Detailed Exploration of Interactive Features

1 Hydration Kinetics & Curing Timeline (Tab 1)

Concrete strength development is non-linear. The Hydration Kinetics simulator recalculates predicted strength across **1, 3, 7, 14, 21, 28, 56, 90, 120, 180, 270, and 365 days** for the exact candidate mix.

- **Active Day Pin:** Highlights the user's selected slider age as a glowing star on the timeline.
- **Early-to-Design Ratio ($f'_{c, 7d} / f'_{c, 28d}$):** Evaluates whether the mix satisfies the industry guideline of achieving 65%–75% of 28-day strength by Day 7.
- **Late-Age Pozzolanic Gain:** Quantifies long-term strength gain at 90 days resulting from secondary C-S-H formation in slag/fly ash mixes.

2 "What-If" Sensitivity Simulator (Tab 2)

Engineers can isolate any constituent (e.g. *Mixing Water*, *Portland Cement*, or *Superplasticizer*) to sweep its physical range across 50 simulated batches while holding all other 7 factors locked. The graph plots the direct response curve and pinpoints the current operating state.

3 Embodied Carbon & Eco-Efficiency (Tab 4)

Using standard cradle-to-gate Life Cycle Assessment (LCA) coefficients ($0.86 ext{ kg CO}_2 ext{e/kg}$ for Portland cement vs 0.07 for slag and 0.015 for fly ash), the app calculates:

- **Total Embodied Carbon:** In $ ext{kg CO}_2 ext{e} / ext{m}^3$.
- **Eco-Efficiency Ratio:** $ ext{MPa} / (100 ext{ kg CO}_2 ext{e})$, quantifying how efficiently the mix converts carbon expenditure into load-bearing capacity.
- **Clinker Replacement Impact:** Direct percentage reduction in carbon emissions compared to an un-blended OPC benchmark.

4 Batch CSV Prediction Engine (Tab 7)

Allows users to drag and drop batches from laboratory trials (e.g. `sample_batch_mixes.csv`). The ML model evaluates hundreds of mixtures in fractions of a second, calculates w/b ratios, assigns structural classifications, visualizes the distribution histogram, and provides an exportable enriched CSV file.

3. Verification & How to Access

Live Server Access

The interactive studio is running live in the background at:

👉 `http://localhost:8501` (or `http://127.0.0.1:8501`)

Scientific Integrity Disclaimer (Mandatory Rule)

Statistical model estimates derived from empirical concrete formulations (Yeh 1998) provide decision support. They do not replace ASTM C39 / BS EN 12390 destructive cylinder testing for certified life-safety structures.